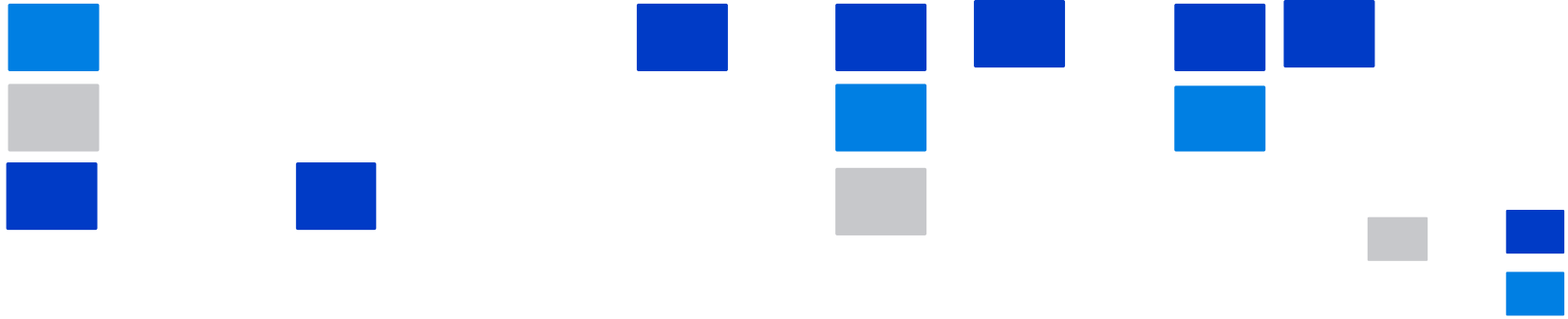




Sistemes Operatius

Tema 4: Crides al sistema - Signals • Curs 2026-27

Signals

Signal: mecanisme del nucli per comunicar esdeveniments als processos.

- Cada signal té associat un nom i un número (**kill -1**).
- **Funcions que podem usar programant:**
 - `signal(int sig, sig_t funció);`
 - `kill(int pid, int sig);`
 - `raise(int sig);`
 - `pause();`
- **Comandes bash interessants:**
 - `kill`
 - `ps`

Example 1

Example 1:

```
void farewell() {
    write(STDOUT_FILENO, "-----\n", 25);
    write(STDOUT_FILENO, "Why do you interrupt me?\n", 25);
    write(STDOUT_FILENO, "-----\n", 25);
    signal(SIGINT, SIG_DFL);
}

int main() {
    signal(SIGINT, farewell);
    while (1) {
        write(STDOUT_FILENO, "Nothing new here.\n", 18);
        pause();
    }
    return 0;
}
```

Exemple 2

Exemple 2:

```
int main() {
    char buf[64];
    int i, n;

    for (i = 1; i <= 64; i++) {
        signal(i, ksighandler);
    }

    alarm(30);

    while (1) {
        n = sprintf(buf, "I am PID %d and I wait for signals\n", getpid());
        write(STDOUT_FILENO, buf, n);
        sleep(2);
    }
}
```

Exemple 2

```
void ksigandler(int signum) {  
    switch (signum) {  
        case SIGINT:  
            kctrlc(signum);  
            break;  
        case SIGALRM:  
            ksigalrm(signum);  
            break;  
        default:  
            kgeneric(signum);  
            break;  
    }  
    signal(signum, ksigandler);  
}
```

Exemple 2

```
void kctrlc(int signum) {  
    char buf[64];  
    int n = sprintf(buf, "I received a ctrl + c (%d).\n", signum);  
    write(STDOUT_FILENO, buf, n);  
}
```

```
void ksigalrm(int signum) {  
    char buf[64];  
    int n = sprintf(buf, "I received a sigalrm (%d).\n", signum);  
    write(STDOUT_FILENO, buf, n);  
}
```

```
void kgeneric(int signum) {  
    char buf[64];  
    int n = sprintf(buf, "I received other signal (%d)\n", signum);  
    write(STDOUT_FILENO, buf, n);  
}
```